

Logic

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P TRUE	Q TRUE	NOT P	P AND Q	P OR Q	(NOT P) OR Q	IF P THEN Q	converse of prev. col.
$\underline{p}$	$\underline{q}$	$\underline{\sim p}$	$\underline{p \wedge q}$	$\underline{p \vee q}$	$\underline{\sim p \vee q}$	$\underline{p \rightarrow q}$	$\underline{q \rightarrow p}$
T	T	F	T	T	T	T	T
T	F	F	F	T	F	F	T
F	T	T	F	T	T	T	F
F	F	T	F	F	T	T	T

Logical Quantifiers and Definitions

- $\forall$: The **universal** quantifier. “For every...” or “for all ...”
- $\exists$: The **existential** quantifier. “There exists...” or “at least one”
- $\exists!$: “There exists a unique ...” “There exists one and only one”
- $\in$: “in”, “is an element of”
- $\mathbb{R}$: The real numbers, or “*the reals*”
- $\mathbb{Z}$: The integers
- $\mathbb{Q}$: Rational numbers
- $a|b$: a divides b.
- **Division:** $a|b$ if and only if $\exists k \in \mathbb{Z}$ such that $ak = b$
- **Even:** z is even if and only if $\exists q \in \mathbb{Z}$ such that $2q = z$
- **Odd:** z is odd if and only if $\exists q \in \mathbb{Z}$ such that $2q + 1 = z$
- **Prime:** We say that a positive integer $p > 1$ is prime if and only if p has no positive factors other than 1 and itself.
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- **Rational:** We say that a real number z is *rational* if and only if there exists integers a and b such that $z = \frac{a}{b}$. If no such pair of integers exists, then we say that z is *irrational*.
 - We denote the set of rational numbers as $\mathbb{Q}$, and so we can write $\mathbb{Q} = \{z \text{ such that } z = \frac{a}{b}, \text{ for some integers } a \text{ and } b\}$.

Three methods of proof for $P \rightarrow Q$:

- 1: Direct Proof
 - Assume P is true... show Q is true
- 2: Prove the contrapositive directly
 - Assume $\sim Q$ is true... show $\sim P$ is true.
- 3: Proof by contradiction
 - Assume $\sim(P \rightarrow Q)$... Arrive at mathematical nonsense/contradiction... Conclude $P \rightarrow Q$

Examples

Example 1

TRUE or FALSE

$\forall x \in \mathbb{R} \exists y \in \mathbb{R}$ such that $y^2 = x$

False Justification: Counterexample, let $x = -5$

Example 2

TRUE or FALSE

$\forall x \in \mathbb{R} \exists y \in \mathbb{R}$ such that $x + y = 0$

TRUE Justification: For every value of x you can find a specific value of y that is its negative. $x = 5, y = -5$ or $x = -12, y = 12$. So, $x = -y$. This is also interesting in that the value of y is unique. The $\exists$ could be replaced with a $\exists!$ here because there is only a single solution.

Example 3

$\exists y \in \mathbb{R} \forall x \in \mathbb{R}$ such that $x + y = 0$

FALSE This can be thought of there being some special y value that works for every single x. There isn't one, this is a totally different statement from the previous one.

Example 4

$\forall x \in \mathbb{R} \exists y \in \mathbb{R}$ such that $x \cdot y = 0$

TRUE. For any x, we can let $y = 0$. If $x = 0$, then y can be any number.

Example 5

$\forall x \in \mathbb{R} \exists! y \in \mathbb{R}$ such that $x \cdot y = 0$

This is different. Remember how it was stated that when $x = 0$ that y can be any number? Well, what that means is that there isn't a single unique value of y that can satisfy this criteria **BECAUSE** we have to have one that works for every possible value of x. $x = 0$ breaks this down, so this is **FALSE**.

Example 6

$\exists y \in \mathbb{R} \forall x \in \mathbb{R}$ such that $x \cdot y = 0$

TRUE $y = 0$ works for every single possible value of x in the real numbers.

Exercise 7

Claim: The sum of an even and an odd is always odd. **Proof:**

Let a be even and b be odd.

Since a is even, $\exists q \in \mathbb{Z}$ such that $a = 2q$. Since b is odd, $\exists w \in \mathbb{Z}$ such that $b = 2w + 1$

Then $a + b = 2q + 2w + 1$

So $2(q + w) + 1$

Since $q + w \in \mathbb{Z}$, $a + b$ is odd. ■

Exercise 8

Claim: Let a, b, c be integers. If $a|b$ and $a|c$ then $a|(b + c)$

Proof:

Assume $a|b$ and $a|c$.

- Then $\exists q \in \mathbb{Z}$ such that $aq = b$
- Also $\exists w \in \mathbb{Z}$ such that $aw = c$

$\therefore b + c = aq + aw = a(q + w)$

Since $q + w \in \mathbb{Z}$, we know that $a|(b + c)$. ■

Exercise 9

Claim: Let a, b, c be integers. If $a|b$ and $a|c$ then $a|(b - c)$

Proof:

Assume $a|b$ and $a|c$.

- Then $\exists q \in \mathbb{Z}$ such that $aq = b$
- Also $\exists w \in \mathbb{Z}$ such that $aw = c$

$\therefore b - c = aq - aw = a(q - w)$

Since $q - w \in \mathbb{Z}$, we know that $a|(b - c)$. ■

Exercise 10

Claim: Let a, b, c be integers. If $a|b$ and $a|c$ then $a|bc$

Proof:

Assume $a|b$ and $a|c$.

- Then $\exists q \in \mathbb{Z}$ such that $aq = b$
- Also $\exists w \in \mathbb{Z}$ such that $aw = c$

$$\therefore bc = aq \cdot aw = a[qaw]$$

Since $qaw \in \mathbb{Z}$, we know that $a|bc$. ■

Exercise 11

Claim:

If $n \in \mathbb{Z}^+$, then $n^2 + n + 3$ is odd.

Proof:

Assume $n \in \mathbb{Z}^+$

Case i: n is even. Then $\exists q \in \mathbb{Z}$ such that $n = 2q$

Then:

$$n^2 + n + 3 = (2q)^2 + (2q) + 3 \quad (1)$$

$$= 4q^2 + 2q + 3 \quad (2)$$

$$= 2(2q^2 + q) + 3 \quad (3)$$

$$= 2(2q^2 + q + 1) + 1 \quad (4)$$

Since $2q^2 + q + 1 \in \mathbb{Z}$, we know that $n^2 + n + 3$ is odd.

case ii. n is odd. Then $\exists w \in \mathbb{Z}$ such that $n = 2w + 1$.

Then:

$$n^2 + n + 3 = (2w + 1)^2 + (2w + 1) + 3 \quad (5)$$

$$= 4w^2 + 4w + 1 + 2w + 1 + 3 \quad (6)$$

$$= 4w^2 + 6w + 4 + 1 \quad (7)$$

$$= 2(2w^2 + 3w + 2) + 1 \quad (8)$$

Since $2w^2 + 3w + 2 \in \mathbb{Z}$, we know that $n^2 + n + 3$ is odd. ■

Exercise 12

Claim:

If m^2 is odd, then m is odd.

m^2 is odd $\rightarrow m$ is odd.

Strategy: Contrapositive: If m is **not** odd, then m^2 is **not** odd.

If m is even then m^2 is even.

Proof:

We prove the contrapositive.

Assume m is even.

Then $\exists q \in \mathbb{Z}$ such that $m = 2q$.

So, $m^2 = (2q)^2 = 2(2q^2)$.

Since $2q^2 \in \mathbb{Z}$, m^2 is even. ■

Exercise 13

If $a|b$ and $a|c$ then $\forall x, y \in \mathbb{Z}, a|(bx + cy)$

Let's break up this sentence. We can think of the portion before "then" as P and the rest as Q. So we have $P \rightarrow Q$.

From here we can break up Q.

Let R be the section $x, y \in \mathbb{Z}$

and S be $a|(bx + cy)$.

So now we can say that $R \rightarrow S$. From there we can rewrite the ENTIRE claim.

$P \rightarrow (R \rightarrow S)$

Our general gameplan here is to assume P, see where that takes us. Then we'll assume R and hope with all of that information combined we can show S.

Proof: Assume $a|b$ and $a|c$. Since $a|b$, $\exists n \in \mathbb{Z}$ such that $b = na$. Also, since $a|c$, $\exists m \in \mathbb{Z}$ such that $c = ma$.

Assume $x, y \in \mathbb{Z}$. Then:

$$bx + cy = nax + may \tag{9}$$

$$= a(nx + my) \tag{10}$$

Since $nx + my \in \mathbb{Z}$, $a|(bx + cy)$. ■

Exercise 14

A famous, long standing (but unproven) conjecture, called the Goldbach Conjecture, states that every even number greater than 2 can be written as the sum of two prime numbers.

(i.e. that there exists prime numbers p_1 and p_2 such that $2 < 2k = p_1 + p_2$, for every $k > 1$)

Prove or disprove: If the conjecture is true, then every odd number greater than 5 can be expressed as the sum of three prime numbers.

If $2k > 2, k \in \mathbb{Z}$, then $2k = p_1 + p_2$.

It is of note that p_n notation represents prime numbers where n just helps differentiate.

Claim: If **Goldbach is true** (P), then every odd number greater than 5 can be expressed as the sum of three primes (Q).

$P \rightarrow Q$

Breaking up Q into an $R \rightarrow S$ gives us:

R: x is an odd number greater than 5

S: x can be expressed as the sum of three primes

$P \rightarrow (R \rightarrow S)$

Structure: Assume P, Assume R and hopefully show S.

We've gotta do something first though. We need a format we can actually work with. Let's use our definition of odd here and manipulate it. We know we're looking at **odd numbers greater than 5**.

$$x = 2j + 1 > 5$$

But this on its own isn't helpful. Goldbachs only works for even numbers greater than 2. We can manipulate our statement though by subtracting by 3.

$$x - 3 = 2j - 2 > 2$$

Now we have a format that does work with goldbachs. We can finish the rewrite here.

$$x - 3 = p_1 + p_2$$

Proof: Assume Goldbach is true. Also assume x is an odd number greater than 5.

Then: $\exists j \in \mathbb{Z}$ such that $x = 2j + 1$

Since $x > 5$, we have $x = 2j + 1 > 5$

Subtracting 3 from each, we have $x - 3 = 2j - 2 > 2$ or $x - 3 = 2(j - 1) > 2$.

In other words, $x - 3$ is an even number greater than 2. Thus, by the Goldbach conjecture, $x - 3$ can be expressed as the sum of two primes. Therefore, $\exists$ prime numbers p_1, p_2 such that:

$$x - 3 = p_1 + p_2 \tag{11}$$

$$x = p_1 + p_2 + 3 \tag{12}$$

Since 3 is prime, x is the sum of three primes. ■

Exercise 15

Claim: The number a is rational iff the number $-a$ is rational.

$$P \longleftrightarrow Q$$

Proof:

($\rightarrow$ only). Assume a is rational.

Then, $\exists m, n \in \mathbb{Z}$ such that $a = \frac{m}{n}; (n \neq 0)$.

Then $-a = -\frac{m}{n} = \frac{-m}{n}$.

Since $-m$ and n are both integers, $-a \in \mathbb{Q}$

($\leftarrow$ only). Assume $-a$ is rational.

$\exists p, q \in \mathbb{Z}$ such that $-a = \frac{p}{q}; (q \neq 0)$.

Then (multiple both sides by -1) $a = -\frac{p}{q} = \frac{-p}{q}$.

Since $-p$ and q are both integers, $a \in \mathbb{Q}$. ■

Exercise 16

Claim: The sum of two rationals is rational.

Let a & b be rational numbers.

Since $a \in \mathbb{Q} \exists c, d \in \mathbb{Z}$ such that $a = \frac{c}{d}; (d \neq 0)$

Since $b \in \mathbb{Q} \exists e, f \in \mathbb{Z}$ such that $b = \frac{e}{f}; (f \neq 0)$

Then $a + b = \frac{c}{d} + \frac{e}{f} = \frac{cf+de}{df}; (df \neq 0)$.

Since $cf + de \in \mathbb{Z}$ and $df \in \mathbb{Z}$, $a + b \in \mathbb{Q}$ ■

Exercise 17

Claim: If x is rational and y is irrational, then $x+y$ is irrational.

$$P \rightarrow Q$$

Note: Proof by contradiction will be used here, so change up the format.

$$\sim (P \rightarrow Q) = P \wedge \sim Q$$

Proof: We shall prove this by contradiction.

Assume x is rational, y is irrational **and** $x + y$ is rational.

Then $\exists a, b, c, d \in \mathbb{Z}$ such that $x = \frac{a}{b}$ and $x + y = \frac{c}{d}$.

Using substitution, we have $x + y = \frac{a}{b} + y = \frac{c}{d}$.

Solving for y , we have $y = \frac{c}{d} - \frac{a}{b} = \frac{cb-ad}{db}$.

Because $cb - ad \in \mathbb{Z}$ and $db \in \mathbb{Z}$, we know $y \in \mathbb{Q}$. However, we assumed above that $y \in \mathbb{Q}$, $\therefore$ we have a contradiction. This allows us to conclude that the sum of a rational and an irrational number is irrational. ■

Exercise 18

Claim: $\forall z \in \mathbb{Q}, \exists x, y \notin \mathbb{Q}$ such that $x + y = z$

Proof:

Let $z \in \mathbb{Q}$.

Then $z + \sqrt{3}$ is irrational, by the previous exercise.

So let $x = z + \sqrt{3}$.

Let $y = -\sqrt{3}$, which is irrational.

Note that $x + y = z + \sqrt{3} - \sqrt{3} = z$. ■

Exercise 19

Claim: $\sqrt{2}$ is irrational.

Proof: (By contradiction)

Assume $\sqrt{2} \in \mathbb{Q}$. Then $\exists a, b \in \mathbb{Z}$ such that $\sqrt{2} = \frac{a}{b}$. We assume $\frac{a}{b}$ is in “lowest terms”, that is, a and b have no common factors.

Since $\sqrt{2} = \frac{a}{b}$, we know $2 = \frac{a^2}{b^2}$.

$\therefore 2b^2 = a^2$. Thus, a^2 is even. So, (by a Lemma that we could prove (will be on the final)), a is even.

Thus $\exists k \in \mathbb{Z}$ such that $a = 2k$.

From here we can state $2b^2 = 2k \cdot 2k = (2k)^2 = 4k^2$.

Since $2b^2 = 4k^2$, $b^2 = 2k^2$. So, b^2 is even, which means (by a Lemma we'll have on the final) b is even. Then, $\exists j \in \mathbb{Z}$ such that $b = 2j$.

So, $\sqrt{2} = \frac{a}{b} = \frac{2k}{2j}$.

Thus, $\frac{a}{b}$ is not in lowest terms as they share a factor of 2. We specified earlier that a and b have no common factors so this is a contradiction. $\therefore \sqrt{2}$ is irrational. ■